

CSSE3113 – Introduction to Software Engineering

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# **Agile Development**

# Outline

## Agile Software Development

- Extreme Programming
- Scrum
- Kanban
- DevOps

# An Agile Process

- Agile means being able to “Deliver quickly. Change quickly. Change often”
- Develops software iteratively
- Delivers multiple ‘software increments’
- Adapts as changes occur

# Agile Software Development

## Characteristics of Agile Software Development

- Small to medium sized teams
- changing requirements
- changing techniques
- Simple design

# Agile Methods

- These methods
  - Focus on the code
  - Are based on an iterative approach to software development
  - Are intended to deliver working software quickly and evolve this quickly to meet changing requirements.
- The aim of agile methods is to reduce overheads in the software process (e.g. by **limiting documentation**) and to be able to respond quickly to changing requirements without excessive rework.

# The Principles Of Agile Methods

| Principle            | Description                                                                                                                                                                         |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Customer involvement | Customers should be closely involved throughout the development process. Their role is provide and prioritize new system requirements and to evaluate the iterations of the system. |
| Incremental delivery | The software is developed in increments with the customer specifying the requirements to be included in each increment.                                                             |
| People not process   | The skills of the development team should be recognized. Team members should be left to develop their own ways of working.                                                          |
| Adapt change         | Expect the system requirements to change and so design the system to accommodate these changes.                                                                                     |
| Maintain simplicity  | Focus on simplicity in both the software being developed and in the development process. Wherever possible, actively work to eliminate complexity from the system.                  |

# Agile Methodologies

1. eXtreme Programming (XP)
2. Scrum
3. Kanban
4. DevOps

# Extreme Programming

- Perhaps the best-known and most widely used agile method.
  - New versions may be built several times per day
  - Increments are delivered to customers every 2 weeks
  - All tests must be run for every build and the build is only accepted if tests run successfully.

# Extreme Programming (XP)

## XP Values

- Communication(Be together with your customer and fellow programmers, and talk to each other )
- Simplicity (Use simple design and programming practices, and simple methods of planning, tracking, and reporting)
- Feedback(Test your program and your practices, using feedback to steer the project)

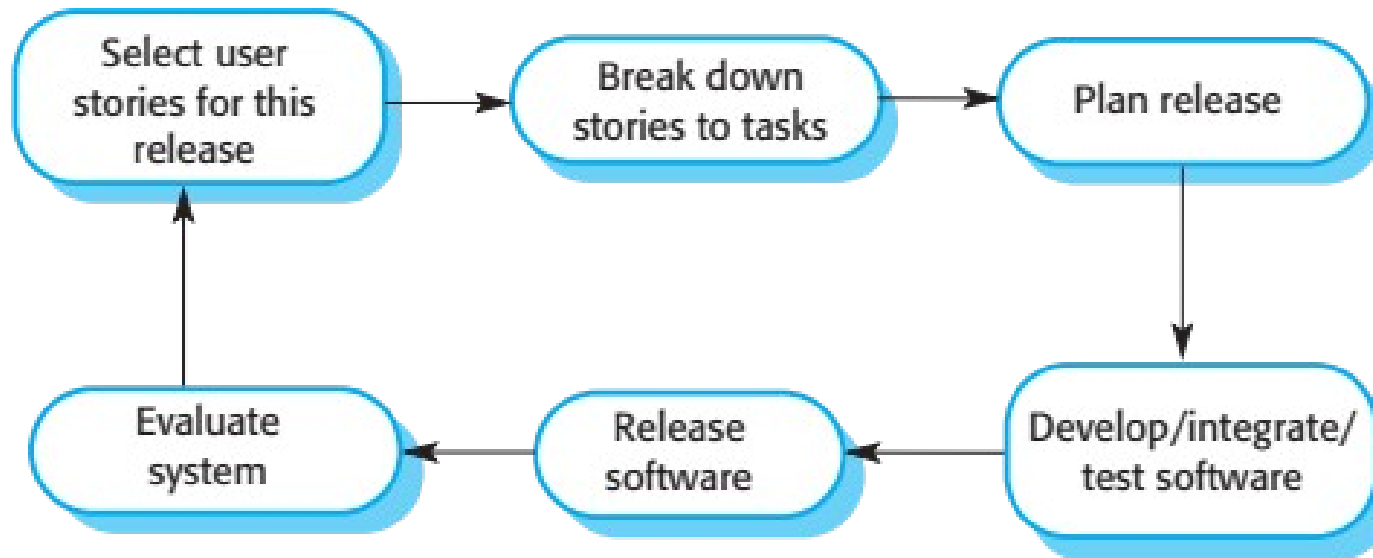
# Extreme Programming Core Practices (A)

| Principle or practice  | Description                                                                                                                                                                               |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Incremental planning   | Requirements are recorded on story cards<br>The developers break these stories into development 'Tasks'.                                                                                  |
| Small releases         | The minimal useful set of functionality that provides business value is developed first.<br>Releases of the system are frequent and incrementally add functionality to the first release. |
| Simple design          | Enough design is carried out to meet the current requirements and no more.                                                                                                                |
| Test-first development | An automated unit test framework is used to write tests for a new piece of functionality before that functionality itself is implemented.                                                 |
| Refactoring            | All developers are expected to refactor / restructure the code continuously as soon as possible code improvements are found.<br>This keeps the code simple and maintainable.              |

# Extreme Programming Core Practices (B)

|                        |                                                                                                                                                                                                                                                                                    |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pair programming       | Developers work in pairs, checking each other's work and providing the support to always do a good job.                                                                                                                                                                            |
| Collective ownership   | The pairs of developers work on all areas of the system, all the developers take responsibility for all of the code. Anyone can change anything.                                                                                                                                   |
| Continuous integration | As soon as the work on a task is complete, it is integrated into the whole system. After any such integration, all the unit tests in the system must pass.                                                                                                                         |
| Sustainable pace       | Large amounts of overtime are not considered acceptable as the net effect is often to reduce code quality and productivity                                                                                                                                                         |
| On-site customer       | <p>A representative of the end-user of the system (the customer) should be available full time.</p> <p>In an extreme programming process, the customer is a member of the development team and is responsible for bringing system requirements to the team for implementation.</p> |

# The Extreme Programming Release Cycle



## Example – Story Card For Downloading Document

### **Downloading and printing an article**

First, you select the article that you want from a displayed list. You then have to tell the system how you will pay for it - this can either be through a subscription, through a company account or by credit card.

After this, you get a copyright form from the system to fill in and, when you have submitted this, the article you want is downloaded onto your computer.

You then choose a printer and a copy of the article is printed. You tell the system if printing has been successful.

If the article is a print-only article, you can't keep the PDF version so it is automatically deleted from your computer.

# Task Cards

## **Task 1: Implement principal workflow**

## **Task 2: Implement article catalog and selection**

## **Task 3: Implement payment collection**

Payment may be made in 3 different ways. The user selects which way they wish to pay. If the user has a library subscription, then they can input the subscriber key which should be checked by the system. Alternatively, they can input an organisational account number. If this is valid, a debit of the cost of the article is posted to this account. Finally, they may input a 16 digit credit card number and expiry date. This should be checked for validity and, if valid a debit is posted to that credit card account.

# What is Scrum?

- Scrum is light weight, interrelated practices and set of laws to optimize the development environment and reduce organizational overheads.
- A Scrum is just like the eight members of the Rugby team, they all are packed together and work as a team. Only one goal has to achieve by the entire team at a single time. [Mike Beedle (2001)]

# Scrum in Detail

## Scrum Phases

There are *three phases* in Scrum given below:-

**Planning** - In planning phase of Scrum model, the requirements are gathered by the customers by scratching a new system

**Development Sprints** - In this phase of Scrum, the developers start coding and developing the functions or module in sprints iterations

**Closure** - In this phase, the final product is released, and documentation is made. Sprint review meeting conduct to check either the goals are achieved or not

# Scrum in Detail.....continue

## Scrum Roles

- **Product Owner** - Product owner has different responsibilities. Their main responsibility is to define the features of the product backlog. Product backlog is the lists of features which are required by the customer
- **Scrum Master** - Scrum Master is the team lead or project manager of the project. It has close links with product owner. It has responsibility to solve all problems faced by the team members during the life cycle
- **Team** - The team members in Scrum consist of seven members. The number can vary from project to project. The main function of the team is to develop the product. The team complete the different tasks assigned by the product owner.

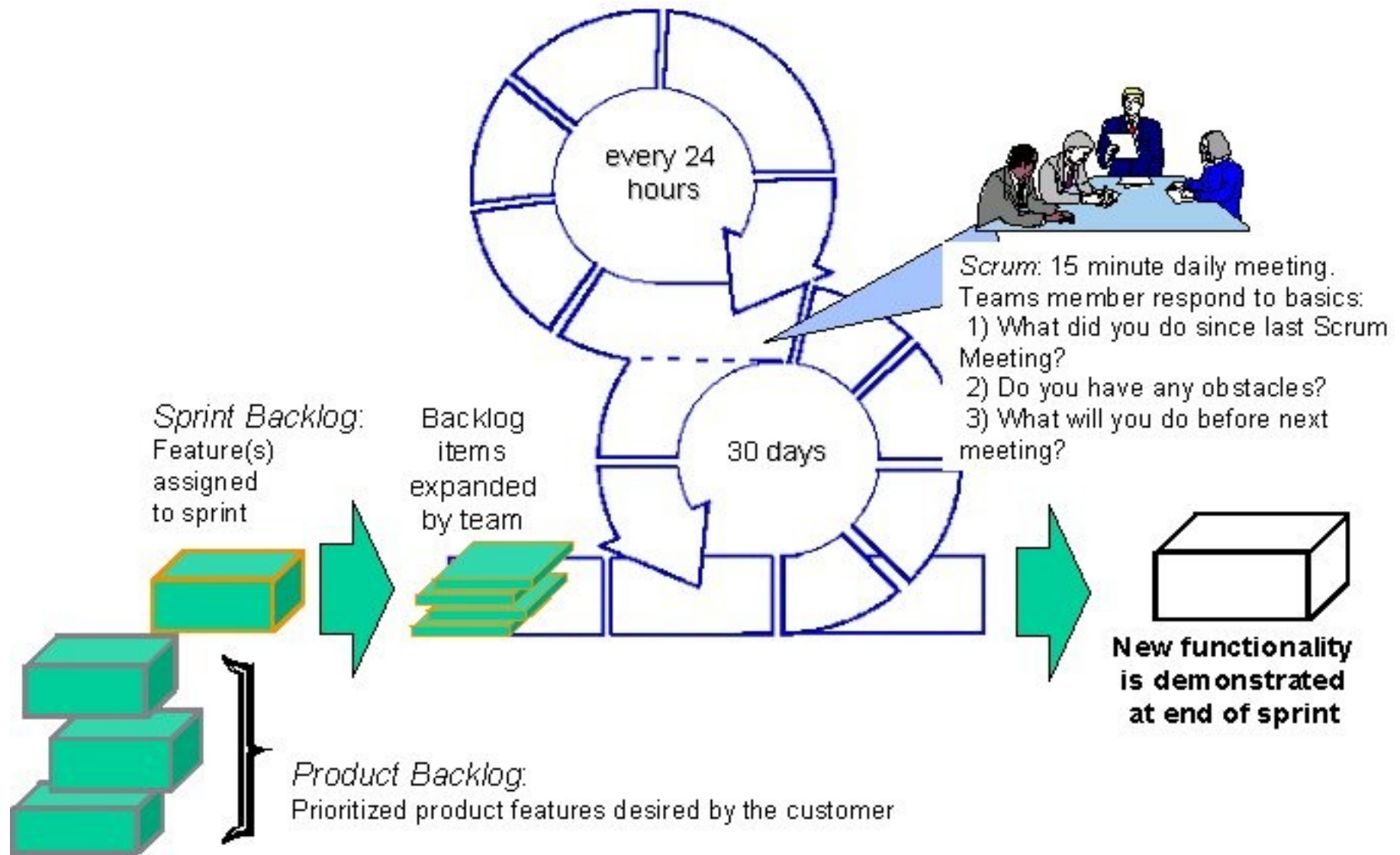
# Scrum in Detail.....continue

## Scrum Artifacts

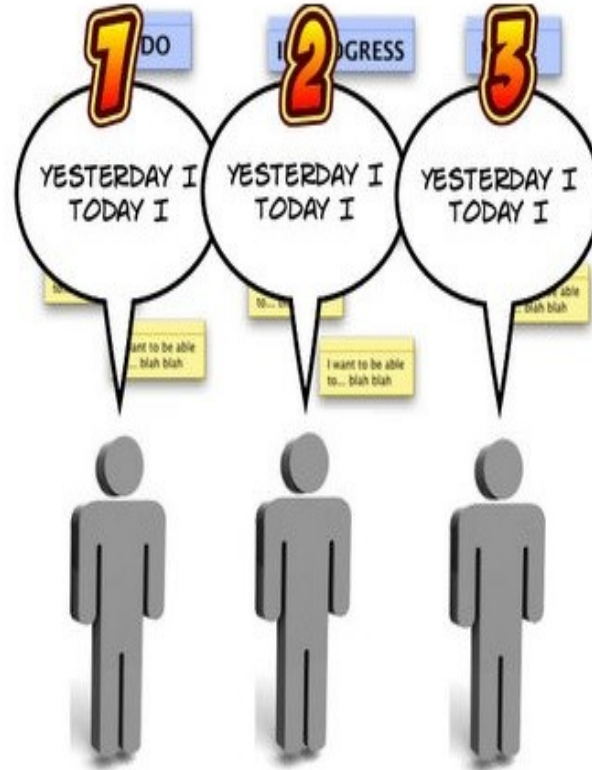
There are three artifacts in Scrum are Product backlog, Sprint Backlog and Burn down Charts.

- **Product Backlog** - In product backlog, product owner prepares a list of customer requirements and then prioritized them with high return of investment. The higher priority function is made first and lower propriety function develops later in development phase.
- **Sprint Backlog** - In sprint backlog, the team of the Scrum is committed to complete all tasks given by the product owner within the time frame. Sprint backlog are maintained by Scrum Master.
- **Burn Down Chart** - In burn down chart sprint progress, release progress and product progress are maintained through graphs. The estimation of the product list is maintained in it. The estimated time are mentioned in the graph to complete the different task. It is maintained by product owner

# Scrum Process Flow



# Standup meetings



# Scrum

- Scrum says that Repeated/Defined problems, solutions, Developers and organizational environment is not entirely possible because developers change from one project to another.
- Scrum assumes that the development process is always empirical and not defined.
- Scrum says uncertainties are impossible to measure, therefore, looks beyond the repeatable /defined approach

# Why Scrum Works

- Most of the defined model assumptions are removed
- Constant feedback
- Focused on “What can be done” instead of “Why it can’t be done”

# Advantages

- ❖ Great emphasis on teamwork
- ❖ Team learns and contributes throughout the process
- ❖ Team becomes autonomous and strives for excellence
- ❖ Rotation of leadership depending on the phase gives a distributed nature of project execution

# Advantages (Cont.)

- ❖ The management team has a pulse on the progress of the team, stepping in whenever required
- ❖ Creates an open environment and encourages feedback
- ❖ Evaluation of effort and subsequent rewards are based on the team performance
- ❖ Reduced need for meetings, authorization and reporting
- ❖ Iterative model leading to a delivery every 30 days

# Disadvantages

- Reusability is not considered in Scrum model.
- In product backlog, the product owner prioritized the item and business values in terms of most return of investment. This will not be suitable in large scale projects.
- Changing requirements is welcome at any stage of the development in Scrum. This will tend to increase the cost of the project and also increase the time.
- In Scrum, a daily meeting takes place for 15 minutes. These meetings are not considered to be a good strategy to cope the problems. Basically, it tends to increase the complexity of the project.
- Increase feedback by the customer will increase the quality of the product but feedback which comes in later stage of the development will become a big problem. It will be very difficult to change the scope of the project in later stage of the development life cycle.

# Kanban

- The word Kanban is of Japanese origin and its meaning is linked to a time concept, “just-in-time”.
- The Kanban method exists in a board or table (**Kanban board**), divided into columns, that **shows every flow of the software production**.
- As the development evolves, the information contained in the table changes, and when a new task comes into play, a new “card” is created.
- The Kanban method requires communication and transparency so that the members of a team can know exactly at what stage the development is and can see the status of the project at any time.

# Kanban

## Advantages

- Ability to view all tasks of one project (Completed, In Progress or In Testing, for example);
- It is possible to limit the number of running tasks (that is, the amount of work, bearing in mind its resolution or deliverability);
- Focus on the duration of a cycle – how long it takes a task to go from backlog to the final stage;
- Allows continuous deliveries.

## Disadvantages

- It is possible for members of a team to misinterpret the information shown on the Kanban Board, especially when it is revealed as outdated;
- Since there are no timeframes in Kanban, you can face time related problems, such as delays, associated with each and every stage.



## PROGRAM BOARD



## PROGRAM BACKLOG

20

| Next PI | PI+1                                                                           | PI+2                                                                   |
|---------|--------------------------------------------------------------------------------|------------------------------------------------------------------------|
|         | Example Feature<br>13                                                          | Enabler15<br>5                                                         |
|         | Infra 21 steps to ensure that we have sound tech in place to support high<br>8 | Feature 15<br>20                                                       |
|         | Feature6<br>30                                                                 | UI to support initial interactions to purchase goods on new site<br>13 |
|         | Feature 14<br>5                                                                | Feature22<br>3                                                         |

## CURRENT PI BACKLOG

1

| Iteration 1                                                        |                                                                       |
|--------------------------------------------------------------------|-----------------------------------------------------------------------|
| High risk item<br>8                                                | Feature8<br>0 OF 5<br>3                                               |
| Iteration 2                                                        |                                                                       |
| Enabler10<br>13                                                    | Explore<br>Enabler to ensure PI+2 can move forward                    |
| Test Enabler<br>Include pre-commit tests in Continuous Integration |                                                                       |
| Iteration 3                                                        |                                                                       |
| Feature7<br>0 OF 4<br>8                                            | Infrastructure<br>Infra. setup for the following systems: Zeta, Delta |
| Explore<br>Number 12 exploration into options for tech             |                                                                       |

## IMPLEMENTING

1

| In Progress                                             | System Demo - ...                                          | Done                                            |
|---------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------|
| FeatureA<br>0 OF 4<br>3                                 | Feature9<br>11 OF 17<br>5                                  | Explore<br>We need to spike out how to...<br>WA |
| Infrastructure<br>Infrastructure enabler<br>21          | FeatureB<br>1 OF 7<br>20                                   | Enabler3<br>13                                  |
| Feature 13<br>1 OF 3<br>8                               | Initial release roundup 1<br>3                             | Enabler 1<br>11 OF 11<br>8                      |
| Architecture<br>Architecture details for iteration<br>5 | The feature that delivers the most value<br>12 OF 12<br>13 | Assess this risk item<br>1                      |

2 OPTIONAL - RELEASING

5 OPTIONAL - LEARNING/FEEDBACK

4 DONE

# DevOps

## DevOps Stages

This approach involves several stages that loop continuously until the desired product exists:-

**Continuous Development** – Software deliverables are broken down and developed in multiple sprints

**Continuous Testing** – Automated testing tools are used to help team members test multiple code increments at the same time.

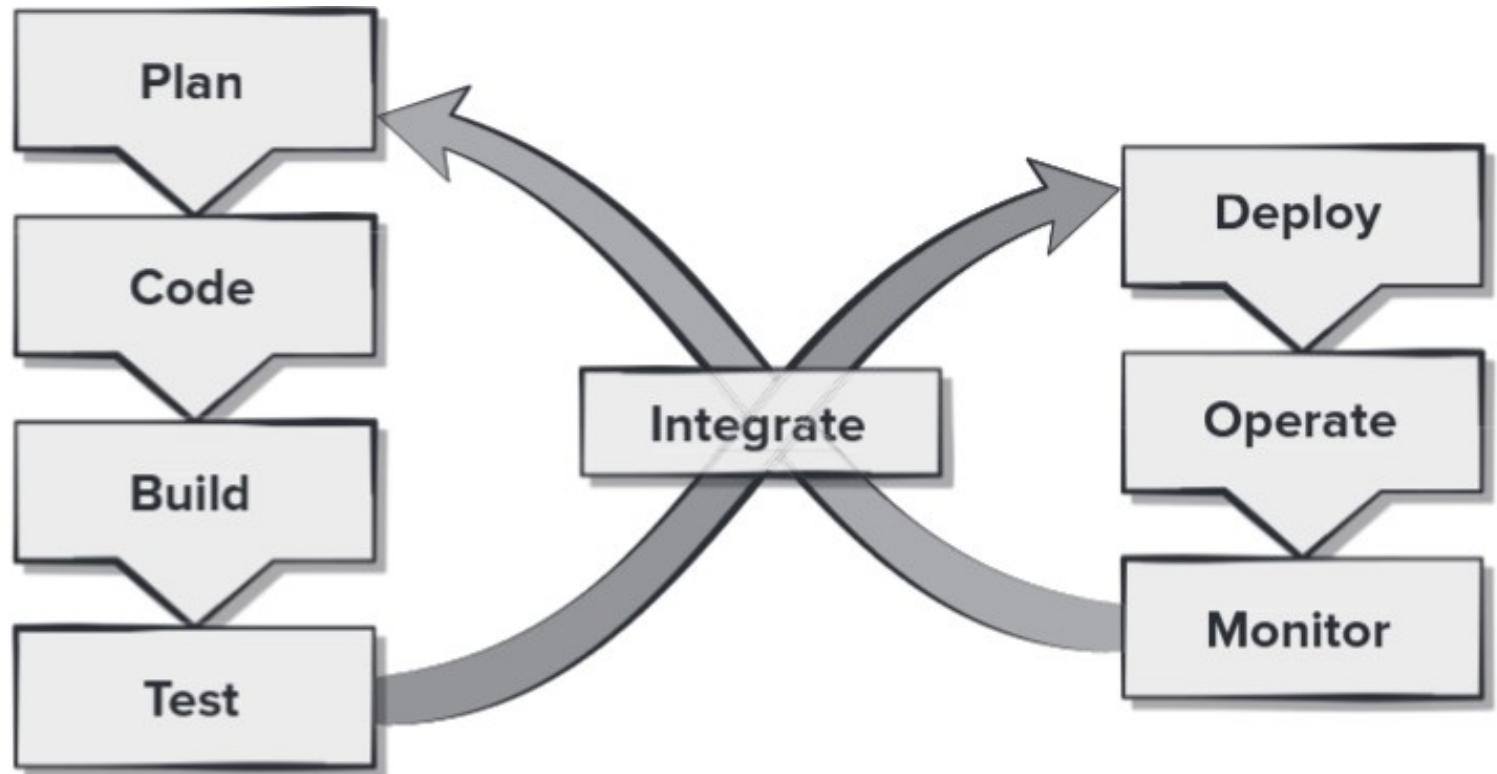
**Continuous Integration** – The code pieces with new functionality are added to the existing code and checked to ensure there are no errors.

**Continuous Deployment** – At this stage the integrated code is deployed to the production environment.

**Continuous Monitoring** – Members in the development team help to improve software quality by monitoring its performance.

# DevOps (Cont.)

DevOps



# Discussion Question

Suggest a suitable process model.

Mughal brothers Pvt Ltd is the distributor of the Qarshi products in Pakistan. This company wants to automate the sale purchase and inventory system. Previously the company has no experience of information technology. They have no idea of what will be the size, cost and duration of the project.